

Software Requirements Specification

<<Annotated Version>>

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<OneStop: Website for Student Project Management>

GROUP 14

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PROJECT

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1.0 Introduction:

OneStop is an online platform which is aimed at helping students manage their projects effectively. OneStop provides tools for scheduling tasks, tracking resources visually with Gantt charts, and developing overall budget plans. This platform enhances the communication and coordination among team members leading to prevention of waste and misunderstanding. The easy-to-use interface offered by OneStop helps students manage their projects in a flexible and efficient manner; thus, it becomes indispensable logistics tool for them.

1.1 Purpose

The OneStop project is aimed at building an online platform that specifically aims to help students do their academic projects in a more efficient way. By providing a full package of tools particularly meant for students.

1.2 Scope of Project

OneStop scope encompasses the student domain, aiming to project tracking by introducing features such as task scheduling and resource management. The core focus lies in developing a seamless tracking platform with precise monitoring, personalize and a user-friendly interface. The project extends to incorporate comprehensive project management suite.

1.3 Glossary

- **Dashboard:** An interface designed to give a quick overview of important project metrics and data, including task status, deadlines, and resource allocation.
- **Gantt Chart:** It is a kind of bar chart that displays the start and end dates of the different project components. It is used to depict a project timeline.
- **Logistics Tool:** This feature help in planning, implementing, and controlling the efficient and effective flow of tasks and resources in a project.
- **Resource Allocation:** This feature will help in assigning and managing resources or tasks in the project.
- **Scheduling:** This will help in planning and organizing task and milestone within the project timeline.
- **Tracking:** The process of monitoring the progress and status of tasks and resources within a project.
- **Collaborators:** This feature help in adding collaborators for the project and teammates can chat with each other.

1.4 References

Wrike and Notion, two well-known applications for project management and organizing, serve as inspiration for OneStop. The robust features and intuitive interfaces of these platforms are well-known, and OneStop functionality and design are modeled after them.

2.0 Overview of document

Requirements:

- Laptop/Desktop
- Connection to the internet

Description:

Onestop is a platform designed specifically for student project management; it includes tools for budget estimation, resource allocation, and work scheduling. The software provides a team chat room, Gantt charts, and real-time communication. It guarantees dynamic tracking, efficient planning, and streamlined management of all project components.

Dashboard:

Experience seamless project management with our intuitive dashboard. Get a comprehensive view of your tasks, progress, resources all in one place. Visualize your project timeline with Gantt charts and monitor key metrics with interactive graphs. Simplify your project management and stay organized with our user-friendly dashboard.

Task Scheduling:

To guarantee that every project milestone is reached on schedule, create, prioritize, and manage tasks with ease. To help your staff stay on task, set deadlines and send out reminders. Streamline your processes and increase output with our intuitive scheduling.

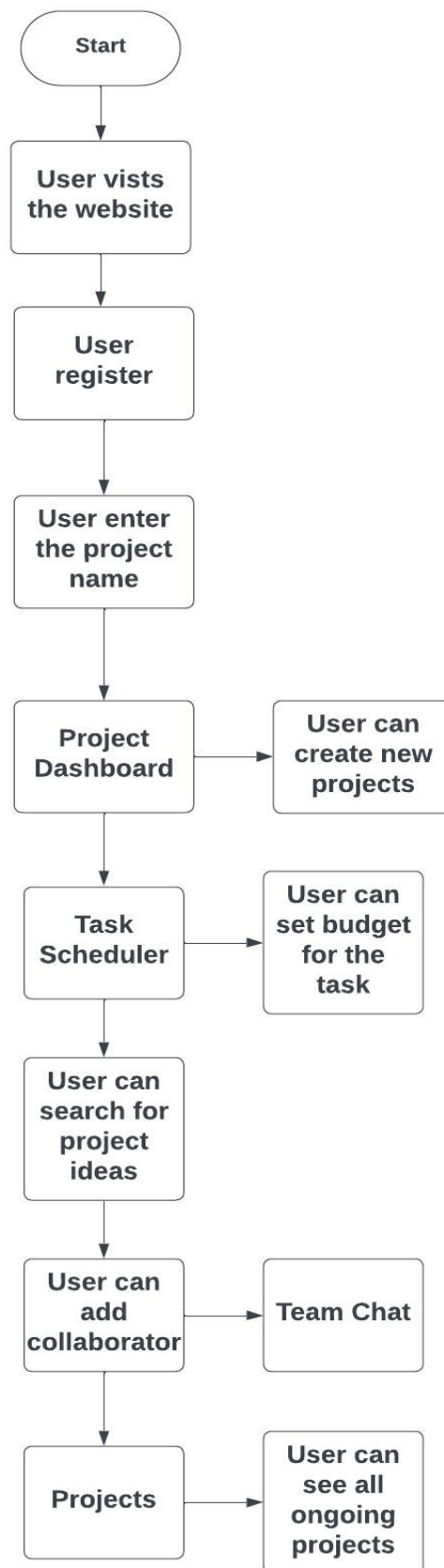
Resource Allocation:

This tool can be used to streamline the way user projects are completed. Make efficient use of team members, time, and supplies to guarantee that all resources are utilized. Monitor the availability and utilization of resources in real-time to prevent shortages or overuse.

Why OneStop?

The aim of this project is to develop a web-based platform called OneStop, specifically designed to address the multifaceted challenges students face when managing projects. OneStop offers a comprehensive suite of tools that simplify and streamline the project management process.

2.1 System Environment



User Perspective:

Welcome to OneStop – The Project Management Hub for students to help them with task scheduling, timely submissions, budget estimation and much more in their project journey.

1. Registration and Profile Creation:

Users start by creating an account on OneStop using their email and phone number.

After registration, users can customize their profiles.

2. Dashboard:

Browse through an extensive dashboard of project, including revenue, progress bar, and notifications for the deadline of the tasks.

Users can watch how much the task impact the progress of the project.

3. Task Scheduling:

Explore a dedicated page for tasks where user can initialize the task with its due date the impact it has for the project like (low, high, moderate) and the budget require for the particular task.

Users can assign the status for the task as completed, in progress or backlog.

4. Collaborators:

User can add collaborator for their project through the email id of another user sending them a invite for accept their request as a collaborator for the project.

5. Team Interaction:

After the adding of collaborator in the project the user can chat with another team members in the website and can discuss the project strategies.

6. Project Ideas:

Receive project recommendations based on preferences.

Stay updated on new project ideas tailored to your interests.

Backend Roles:

The backend of OneStop plays a crucial role in ensuring a seamless and efficient user experience. Here are some key backend responsibilities:

Backend Roles and Responsibilities

1. Node.js Developer

- **Server Setup and Maintenance:** Setting up the server environment using Node.js and ensuring it runs smoothly.
- **API Development:** Creating and managing RESTful APIs for data interaction between the frontend and backend.
- **Database Integration:** Connecting the application with MySQL, managing CRUD operations, and ensuring efficient data retrieval and storage.
- **Authentication and Authorization:** Implementing secure user authentication and role-based access control.

2. Express.js Developer

- **Routing Management:** Managing application routing and handling different HTTP requests (GET, POST, PUT, DELETE).
- **Middleware Integration:** Implementing and managing middleware functions for request processing.
- **Error Handling:** Developing robust error handling mechanisms to ensure smooth application performance.

3. MySQL Database Administrator

- **Database Design and Management:** Designing the database schema, creating tables, indexes, and ensuring data integrity.
- **Query Optimization:** Writing optimized SQL queries to improve database performance.
- **Backup and Recovery:** Implementing regular backup procedures and ensuring data recovery options are in place.

4. Collaboration and Communication Tools Developer

- **Team Chat Room:** Creating real-time communication tools for team members to discuss projects.
- **Notification System:** Implementing notifications for task updates, deadlines, and other important events.

5. Security Specialist

- **Data Security:** Ensuring all data interactions are secure, encrypting sensitive information, and protecting against SQL injection, cross-site scripting (XSS), and other vulnerabilities.
- **User Authentication:** Implementing secure login systems using techniques like JWT (JSON Web Tokens) or OAuth.

6. Integration and Deployment Specialist

- **Continuous Integration and Deployment (CI/CD):** Setting up CI/CD pipelines for automated testing and deployment.
- **Cloud Integration:** Integrating the application with cloud services (e.g., AWS, Azure) for scalability and reliability.

List of cases where OneStop is used:

- **Group Projects and Assignments:** Enables collaboration and task management among students involved in group projects or assignments.
- **Capstone Projects:** Assist students in their final year in effectively planning, managing, and executing their capstone projects.
- **Research Projects:** Help in organizing and managing research activities, timelines, resources, and budgets.
- **Event Planning:** Allowing students to organize and run campus events, whether workshops, seminars, or cultural, effectively scheduling tasks and allocating resources.
- **Club and Society Activities:** Structuring the running of activities, projects, and events by student clubs and societies.
- **Personal Projects:** Help students manage personal projects, such as building a portfolio, creating a blog, or even just a side project.
- **Collaborative Learning:** Manage collaborative learning projects in which students learn by working together to achieve common learning objectives.
- **Start-up Initiatives:** Guide student entrepreneurs in the planning and management of their start-up projects from ideation to execution.
- **Community Service Projects:** Guide students in designing the community service projects and carry them out to ensure the time and resources are used efficiently.
- **Extracurricular Activities:** Execute extracurricular projects, activities, and functions like theatre/drama productions, exhibitions of artwork, or sports events.

Website making details:

OneStop Tracking Flow:

Project Tracking Flow

1. Project Initialization

User Registration and Authentication:

User registration and authentication module will be developed.

Dashboard Overview:

Design an easy-to-use dashboard to represent an overview of all projects.

2. Create and Manage Projects

Create New Project:

The facility to create a new project, specifying project name, description, and other information, such as deadlines.

Project Dashboard:

A dedicated dashboard for every project showing crucial metrics and progress.

3. Task Scheduler and Management

Task Creation:

Create a task and assign it; assign tasks by specifying details such as due dates, priority levels, and task descriptions.

Task Assignment:

The facility to add collaborators and give certain tasks to team members

Task Prioritization:

Provide features for prioritizing tasks and setting deadlines

4. Visual Representation Tools

Gantt Charts:

Interactive Gantt charts of project timelines with task dependencies

Bar Graphs and Line Graphs:

Visualization of project progress, the completion rate of tasks, and many more.

Resource Allocations:

Allow the user to assign time, material, and team members to a specific task.

Resource Tracking:

Track usage and availability of resources.

5. Collaboration Tools**Team Chat Room:**

Provide real-time chat functionality for team communication.

Document Sharing:

This would allow users to share documents about the project.

6. Project Tracking and Reporting**Tracking Progress:**

Track the progress of tasks and the whole project.

Generate Reports:

Reports on project status, task completion, used resources, and budget.

3.2 Functional Requirements:**User Authentication and Authorization**

Account Creation and Management

The system shall provide the ability for a user to create an account using their email and password.

The system shall support secure login and logout functionality with password recovery.

Role-Based Access Control

Users shall have pre-assigned roles, such as a Student or Team Leader, that allow them to perform various activities.

Recover Password and Reset

The system shall allow users to recover their password through a secure password reset process

A reset link shall be sent via the email registered by the user.

3.2.1 Dashboard View**Project Central Dashboard**

The dashboard shall offer a central view of all the projects at hand.

It should indicate upcoming deadlines and task status through visual expressions.

Task Overview and Summary

It shall provide a summary of tasks, showing the status, priority, and deadlines.

There should be an overdue and an upcoming summary of tasks.

Key Performance Indicators

The dashboard shall include key performance indicators for the progress of the project, resource usage, and budget status.

Task Management

Task Creation and Editing

Users can create new tasks with details of the title, description, priority, and deadline.

Users should be able to edit and update details of tasks according to requirement.

Task Assignment and Scheduling

Tasks can be assigned to team members. There should be options for due dates and checking the progress of the same.

There should be a graphical drag and drop task scheduler to organize tasks.

Notification and Reminders

Users will be informed or reminded of oncoming deadlines and task changes.

Configurable alert preferences for different types of notifications.

3.2.2 Collaborator Management

Adding and Removing Collaborators

Facility to add, remove and manage collaborators on a project.

Viewing and editing collaborator roles and permissions.

3.2.3 Visual Representations

Graphical Displays

Display of bar graphs and line graphs for tracking progress on the tasks and usage of resources.

3.2.4 Resource Management

Resource Allocation

Resource allocation and tracking tool for time material resources and budget.

Reports of resource usage and warnings in case resources will run short.

3.2.5 In-App Chat Room

In-app chat room for discussion and communication among the team.

3.2.6 Project Tracking

Dynamic Tracking

Dynamic project tracking wherein the status of the project and progress will get updated.

3.2.7 Categorization and Filtering

Facility to categorize and filter projects on different criteria like by deadline, by priority.

3.2.8 User Interface and Experience

Interface Design

Intuitive User Interface design.

Responsive Design

Responsive design to ensure accessibility on various devices: desktop, tablet, and mobile.

Data Management

3.2.9 Reporting and Analytics

Report Generation

Generate project reports regarding task completion, resource utilization, and budget status.

Analytics Dashboard

Analytics dashboard to track key performance indicators (KPIs).

3.2.10 Integration Capabilities

Third-Party Integrations

Option to integrate with third-party tools or services—like calendar applications.

3.2.11 Performance and Security

Performance

Fast loading and processing of data.

Security

Put in place security features that ensure the protection of users' and projects' data.

3.2.12 Backend Administration:

User Management:

Admin panel for managing user accounts and roles.

3.2.13 Search:

Search Functionality:

To search about project ideas.

3.2.14 Security and Privacy:

Authentication Security:

Secure user authentication and session management.

HTTPS protocol for secure data transfer.

Privacy Settings:

User-controlled privacy settings for profile visibility.

Data encryption for sensitive information.

3.3 Non-Functional Requirements:

3.3.1. Performance:

Response Time:

The website should load within 2-3 seconds to ensure a smooth user experience.

Scalability:

The system should handle a scalable number of users..

3.3.2. Security:

Data Encryption:

Ensure all sensitive user data, including passwords and payment information, is encrypted during transmission.

Authentication and Authorization:

Implement secure authentication mechanisms and role-based access controls to protect user accounts and data.

3.3.3. Scalability:

Database Scalability:

The backend database should scale to accommodate a growing volume of data.

Server Scalability:

The server infrastructure should scale horizontally to handle increased user load.

3.3.4. Usability:

User Interface Consistency:

Maintain a consistent and intuitive user interface across all pages of the website.

Accessibility:

Ensure the website is accessible to users with disabilities, following accessibility standards.

3.3.5. Compatibility:

Browser Compatibility:

The website should be compatible with popular browsers such as Chrome, Firefox, Safari, and Edge.

3.3.6. Maintainability:

Code Maintainability:

Write clean, modular, and well-documented code to facilitate easy maintenance.

Frontend: HTML5, CSS3, JavaScript.

Backend: Node.js, Express.js.

Database: MySQL.

Use Restful APIs and other integrations.